Robust Regression and Outlier Detection

Chapter 2: Simple Regression

Peter J. Rousseeuw

Annick M. Leroy

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Example 4: Brain and Weight Data

Table 7 presents the brain weight (in grams) and the body weight (in kilograms) of 28 animals. [1], [2, .] It is to be investigated whether a larger brain is required to govern a heavier body. A clear picture of the relationship between the logarithms (to the base 10) of these measurements is shown in Figure 7. This logarithmic

- placeholder for the table -
- placeholder for the plot -

transformation was necessary because plotting the original measurements would fail to represent either the smaller or the larger measurements. Indeed, both original variables range over several orders of magnitude. A linear fit to this transformed data would be equivalent to a relationship of the form

$$\hat{y} = \hat{\theta}'_2 x^{\hat{\theta}_1}$$

between brain weight (y) and body weight (x). Looking at Figure 7, it seems that this transformation makes things more linear. Another impor- tant advantage of the log scale is that the heteroscedasticity disappears. The LS fit is given by

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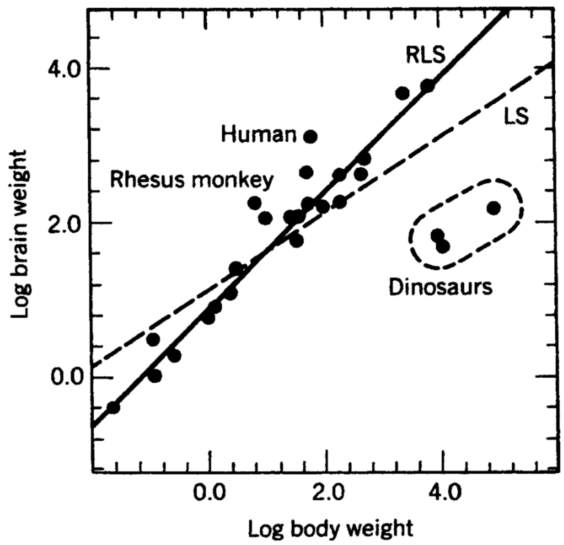

Figure 1: Logarithmic brain weight versus logarithmic body weight for 28 animals with LS (dashed line) and RLS fit (solid line).

(dashed line in Figure 1). The standard error associated with the slope equals 0.0782, and that of the intercept term is 0.1794. In Section 3, we explained how to construct a confidence interval for the unknown regres- sion parameters. For the present example, n = 28 and p = 2, so one has to use the 97.5% quantile of the t-distribution with 26 degrees of freedom, which equals 2.0555. Using the LS results, a 95% confidence interval for the slope is given by t0.3353; 0.65673. The RLS yields the solid line in Table 1, which is a fit with a steeper slope:

Table 1: Brain and body weights of 28 animals.

	index	Species	brain	body
0	1	Mountain beaver	8.1	1.350
1	2	Cow	423.0	465.000
2	3	Grey wolf	119.5	36.330
3	4	Goat	115.0	27.660
4	5	Guinea pig	5.5	1.040
5	6	Dipliodocus	50.0	11700.000
6	7	Asian elephant	4603.0	2547.000
7	8	Donkey	419.0	187.100
8	9	Horse	655.0	521.000
9	10	Potar monkey	115.0	10.000
10	11	Cat	25.6	3.300
11	12	Giraffe	680.0	529.000
12	13	Gorilla	406.0	207.000
13	14	Human	1320.0	62.000
14	15	African elephant	5712.0	6654.000
15	16	Triceratops	70.0	9400.000
16	17	Rhesus monkey	179.0	6.800
17	18	Kangaroo	56.0	35.000
18	19	Golden hamster	1.0	0.120
19	20	Mouse	0.4	0.023
20	21	Rabbit	12.1	2.500
21	22	Sheep	175.0	55.500
22	23	Jaguar	157.0	100.000
23	24	Chimpanzee	440.0	52.160
24	25	Rat	1.9	0.280
25	26	Brachiosaurus	154.5	87000.000
26	27	Mole	3.0	0.122
27	28	Pig	180.0	192.000

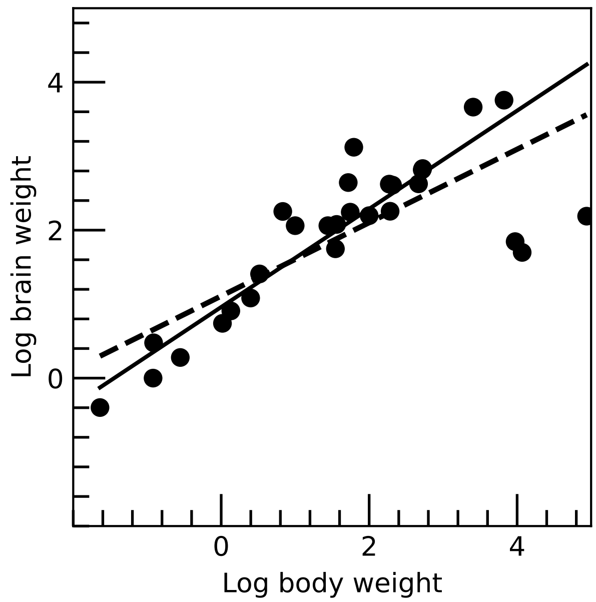

Figure 2: Logarithmic brain weight versus logarithmic body weight for 28 animals with LS (dashed line) and RLS fit (solid line).

Bibliography

[1] S. Weisberg, "Some large-sample tests for nonnormality in the linear regression model: Comment," *Journal of the American Statistical Association*, vol. 75, no. 369, pp. 28–31, 1980.

[2] H. J. Jerison, *Evolution of the brain and intelligence*. Academic Press, 1973.